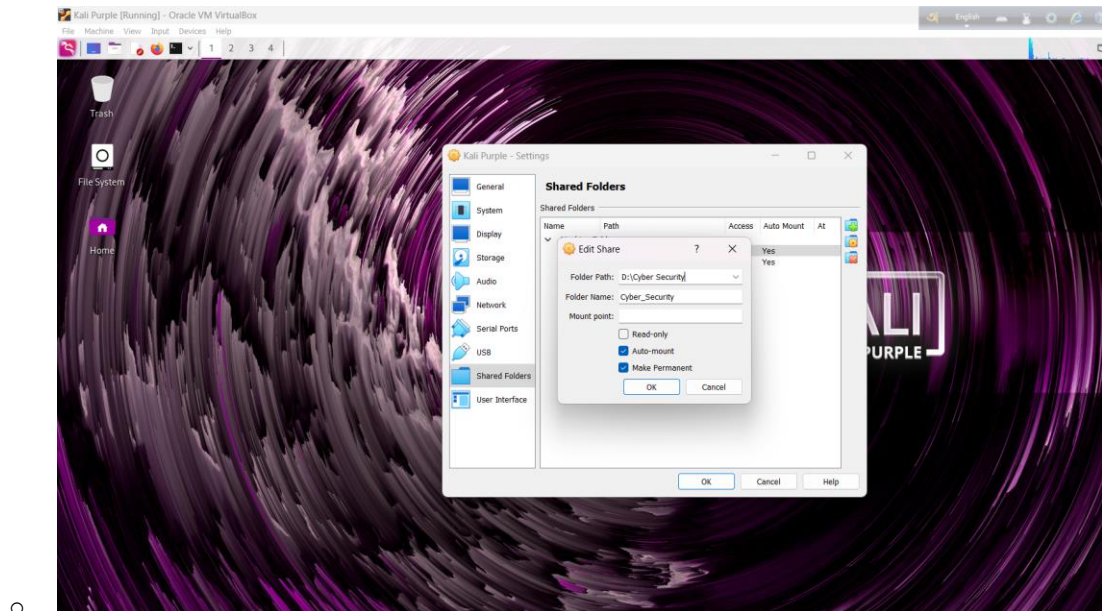


Cyber Security:Class-05
Date: 21 September 2024
Topics : Forensic of an Image

Steps:

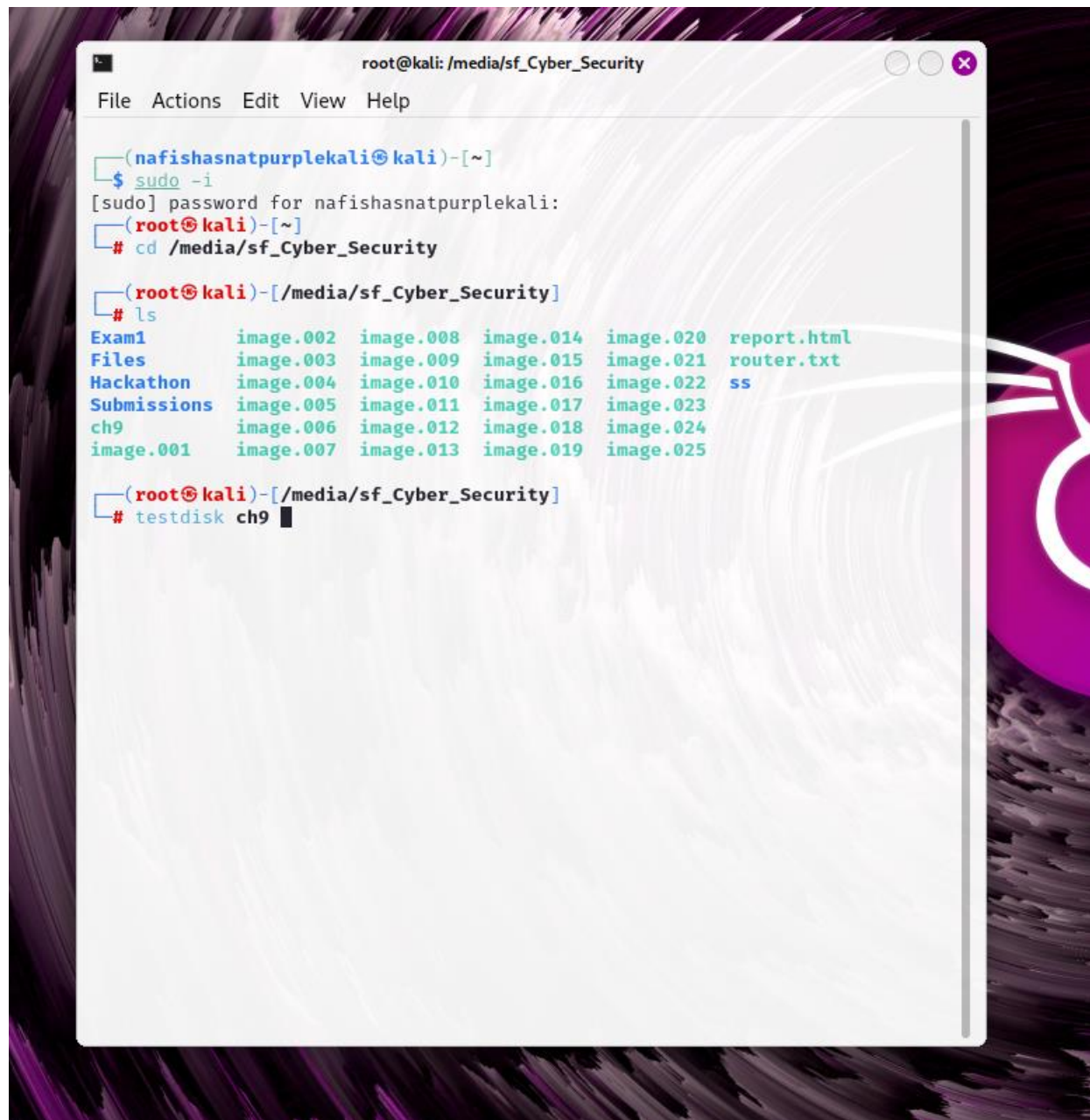
1. Configure the Shared Folder:

- Navigate to **Settings** → **Shared Folders** → **Add Shared**.
- Under **Folder Path**, select **Other**.
- Choose the desired folder to share.



2. Verify Shared Folder in Terminal:

- Gain root access by typing: `sudo -i`.
- Navigate to the shared folder using: `cd /media/sf_JU` (*sf* represents "shared folder," and *JU* is the folder name).
- Confirm that the target file exists in the shared folder.

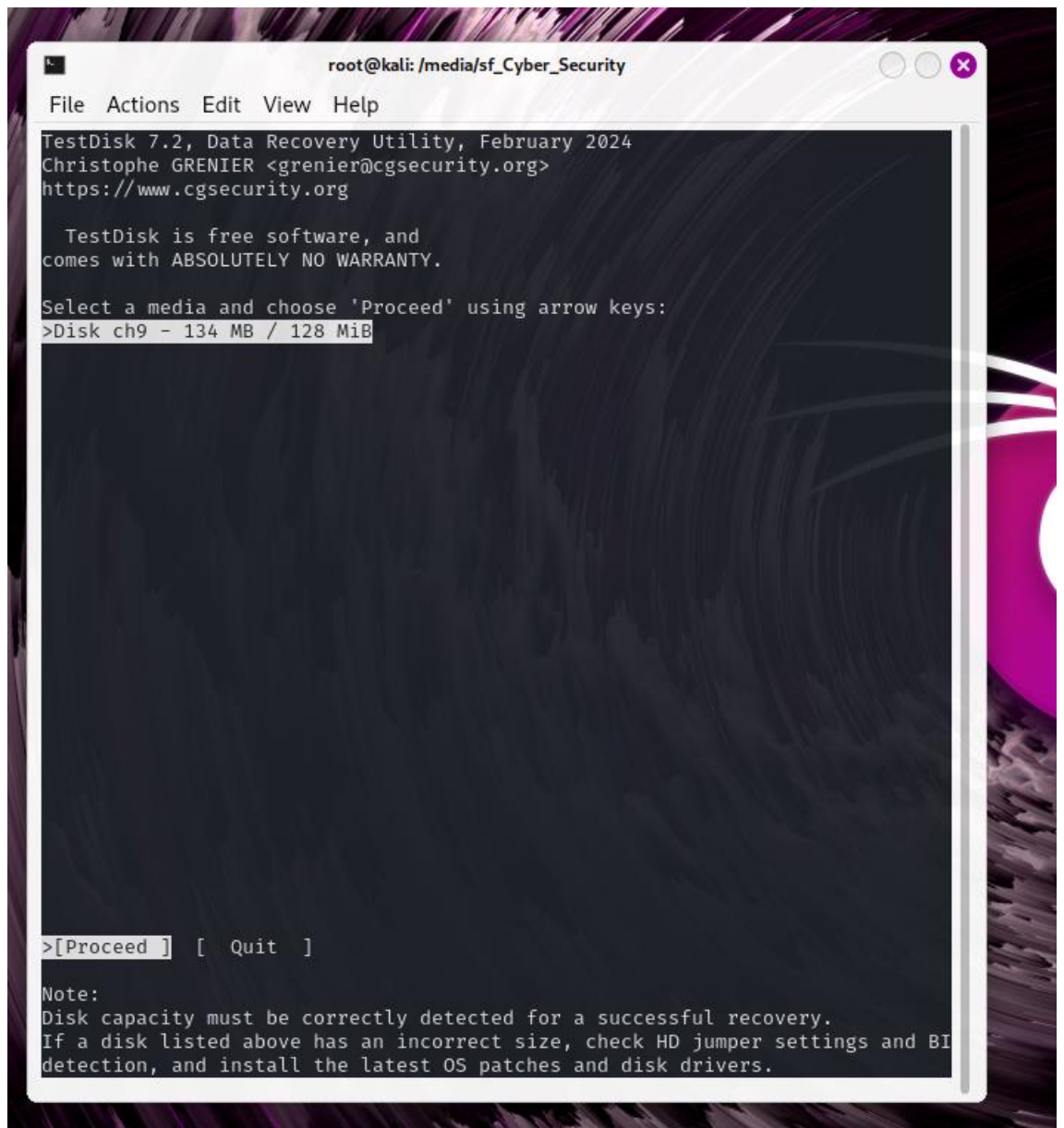


A terminal window titled 'root@kali: /media/sf_Cyber_Security' with a menu bar (File, Actions, Edit, View, Help). The terminal shows the following commands and output:

```
(nafishasnatpurplekali@kali)~  
$ sudo -i  
[sudo] password for nafishasnatpurplekali:  
(root@kali)~  
# cd /media/sf_Cyber_Security  
  
(root@kali)-[/media/sf_Cyber_Security]  
# ls  
Exam1      image.002  image.008  image.014  image.020  report.html  
Files      image.003  image.009  image.015  image.021  router.txt  
Hackathon  image.004  image.010  image.016  image.022  ss  
Submissions image.005  image.011  image.017  image.023  
ch9        image.006  image.012  image.018  image.024  
image.001  image.007  image.013  image.019  image.025  
  
(root@kali)-[/media/sf_Cyber_Security]  
# testdisk ch9
```

3. Recover Deleted Files:

- Use the command `testdisk ch9`, and follow these steps:
 - Select **Process** → **Intel** → **Advanced** → **Boot** → **List** → **Files**.
 - Locate the red line, navigate to it, and type **C** twice to copy the deleted file to the target folder (*JU*).



```
root@kali: /media/sf_Cyber_Security

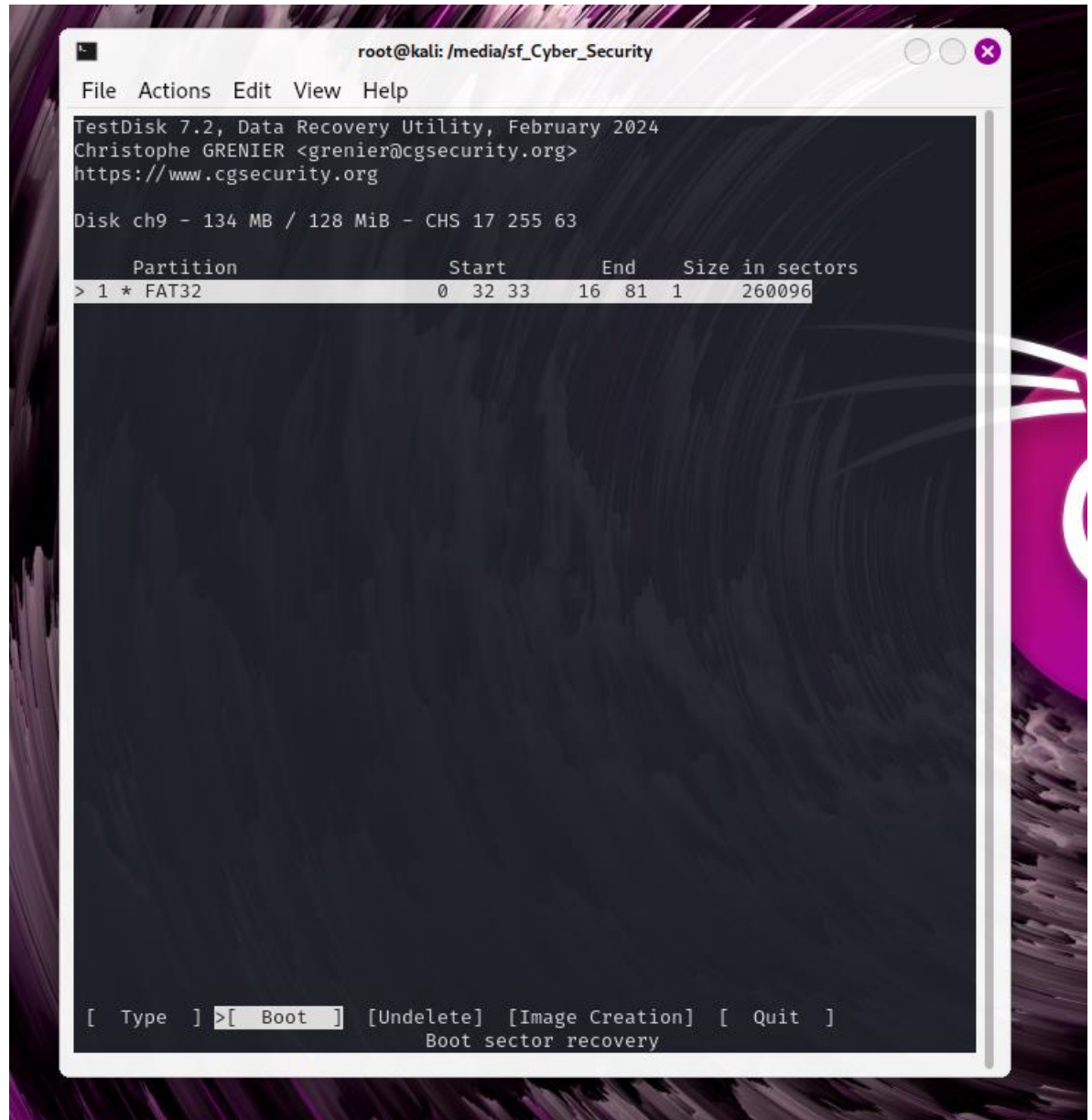
File  Actions  Edit  View  Help

TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk ch9 - 134 MB / 128 MiB
    CHS 17 255 63 - sector size=512

[ Analyse ] Analyse current partition structure and search for lost partiti
>[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ MBR Code ] Write TestDisk MBR code to first sector
[ Delete ] Delete all data in the partition table
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatche
d.
```

4. Extract Files:

- Open the target folder (*JU*), where a new folder named "File" will be created.
- Change the file extension to ZIP (without compressing).
- Extract all files from the ZIP folder.

5. Analyze Picture Metadata:

- Navigate to the folder containing the pictures using `cd`.
- Use the command: `exiftool picture_name` to view detailed metadata of the image.

```
(root@kali)-[/media/sf_CS_Practise]
# exiftool IMG_20231228_125543.jpg
ExifTool Version Number      : 12.76
File Name                    : IMG_20231228_125543.jpg
Directory                    : .
File Size                    : 4.5 MB
File Modification Date/Time   : 2024:08:30 11:34:55-04:00
File Access Date/Time        : 2024:12:03 13:15:27-05:00
File Inode Change Date/Time   : 2024:08:30 11:34:55-04:00
File Permissions              : -rwxrwx---
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order               : Big-endian (Motorola, MM)
Y Resolution                  : 72
X Resolution                  : 72
Camera Model Name             : KB2005
Make                         : OnePlus
Y Cb Cr Positioning           : Centered
Exif Version                  : 0220
Aperture Value                : 1.7
Scene Type                    : Unknown (0)
Exposure Compensation         : 0
Exposure Program              : Not Defined
Color Space                   : sRGB
Max Aperture Value            : 1.7
Exif Image Height             : 3000
Brightness Value              : 0
Date/Time Original            : 2023:12:28 12:55:44
Flashpix Version              : 0100
Sub Sec Time Original          : 085083
White Balance                 : Auto
Interoperability Index        : R98 - DCF basic file (sRGB)
Exposure Mode                 : Auto
Exposure Time                 : 1/177
Flash                         : Off, Did not fire
Sub Sec Time                  : 085083
F Number                      : 1.8
ISO                           : 100
Exif Image Width              : 4000
Components Configuration      : Y, Cb, Cr, -
Focal Length In 35mm Format    : 0 mm
Sub Sec Time Digitized         : 085083
Create Date                   : 2023:12:28 12:55:44
Shutter Speed Value           : 1/177
Metering Mode                 : Center-weighted average
Focal Length                   : 4.7 mm
Scene Capture Type            : Standard
Light Source                   : D65
Sensing Method                : Unknown (0)
Orientation                   : Horizontal (normal)
Resolution Unit                : inches
```

6. Locate the Picture:

- Obtain the GPS Latitude and Longitude from the metadata.
- Visit [GPS Coordinates](#) and input the coordinates to determine the location.

```
File Actions Edit View Help
Focal Length In 35mm Format : 0 mm
Sub Sec Time Digitized : 085083
Create Date : 2023:12:28 12:55:44
Shutter Speed Value : 1/177
Metering Mode : Center-weighted average
Focal Length : 4.7 mm
Scene Capture Type : Standard
Light Source : D65
Sensing Method : Unknown (0)
Orientation : Horizontal (normal)
Resolution Unit : inches
Modify Date : 2023:12:28 12:55:44
GPS Latitude Ref : Unknown ( )
GPS Altitude Ref : Above Sea Level
GPS Processing Method :
GPS Longitude Ref : Unknown ( )
GPS Time Stamp : 00:00:00
GPS Date Stamp :
Compression : JPEG (old-style)
Thumbnail Offset : 1083
Thumbnail Length : 48588
XMP Toolkit : Adobe XMP Core 5.1.0-jc003
Capture Mode : Photo
Lens Facing : Back
Scene Detect Result Ids : [0, 0, 0]
Scene Detect Result Confidences : [0.0, 0.0, 0.0]
Is HDR Active : False
Is Night Mode Active : False
Is Bokeh Active : False
Warning : FPXR segment too small
Image Width : 4000
Image Height : 3000
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Aperture : 1.8
Image Size : 4000x3000
Megapixels : 12.0
Shutter Speed : 1/177
Create Date : 2023:12:28 12:55:44.085083
Date/Time Original : 2023:12:28 12:55:44.085083
Modify Date : 2023:12:28 12:55:44.085083
Thumbnail Image : (Binary data 48588 bytes, use -b option to extract)
GPS Date/Time : 00:00:00Z
GPS Latitude :
GPS Longitude :
Focal Length : 4.7 mm
Light Value : 9.1

(root@kali)-[/media/sf_CS_Practise]
# █
```

This assignment demonstrates how to set up a shared folder, recover and analyze files, and extract location data from images.